

Cell Structure & Intracellular Organelles: An Overview - Factoid Sheet

Week 2 Cellular & Molecular Foundations • Core 01 • memorize-first companion

Core frame: know what each organelle does, how membrane asymmetry is maintained, how passive vs active transport differ, and what the Na⁺/K⁺-ATPase gradients power.

Cell + nucleus - memorize cold

- **Cytosol:** ~pH 7.2; extracellular fluid ~pH 7.4.
- **Only nucleus + nucleolus** are visible by light microscopy in this handout; most other organelles require EM.
- **Malignant cells:** often increased nuclear:cytoplasmic ratio.
- **Nucleosome:** DNA around 8 histones - 2 each H2A, H2B, H3, H4.
- **Euchromatin:** less condensed, accessible, transcriptionally active.
- **Heterochromatin:** condensed, inaccessible, transcriptionally silent.
- **Nucleolus:** rRNA transcription, pre-rRNA processing, ribosomal-subunit assembly.

Chromatin direction

Histone acetylation + DNA demethylation favor euchromatin.
DNA methylation + histone deacetylation favor heterochromatin.
Cells with high protein-synthesis/proliferative activity may have more prominent nucleoli.

Organelle one-liners

Organelle	Must know
Mitochondria	Aerobic ATP production; inner-membrane cristae; matrix ~pH 8.0; own circular DNA + ribosomes.
RER	Production/processing of membrane-embedded and secreted proteins; ER lumen is oxidizing and Ca ²⁺ -rich.
SER	Lipid metabolism, drug/poison detoxification, Ca ²⁺ storage; sarcoplasmic reticulum is specialized SER.
Golgi	Cis receives from ER; trans ships. Glycosylation, sulfation, phosphorylation, limited proteolysis, sorting/packaging.
Lysosome	Acid hydrolases; lumen ~pH 4.7; degrades endocytosed/autophagic material and some pathogens.
Peroxisome	VLCFA beta-oxidation, branched-chain alpha-oxidation, H ₂ O ₂ generation + catalase detoxification.
Ribosome	Protein synthesis. Prokaryote 70S = 30S+50S; eukaryote 80S = 40S+60S.
Proteasome	Polyubiquitin-tagged endogenous proteins -> ATP-dependent unfolding -> peptide degradation.
Centrosome/centriole	Microtubule organization and mitotic spindle; centriole = 9 triplet microtubules and can form basal bodies.

Mitochondrial inheritance

- **mtDNA:** ~16.5 kb, 37 genes.
- Encodes **13 proteins, 22 tRNAs, 2 rRNAs**.
- Maternal transmission of mtDNA disease.
- Affected males do not transmit mtDNA disease.
- **Heteroplasmy** explains variable severity among tissues/children.

Mature RBC exception

Circulating fully differentiated RBCs have proteasomes but no other organelles in this handout.
Clinical tie-ins shown: hereditary spherocytosis (spectrin/peripheral cytoskeleton), proteasome inhibition with bortezomib.

Membrane asymmetry

Location / protein	Factoid
Inner leaflet	PS, PE, PI are enriched on the cytosolic side.
Outer leaflet	PC, sphingomyelin, and glycolipids are enriched externally.
Flippase	Energy-dependent inward movement of selected phospholipids; helps establish/maintain asymmetry.
Floppase	ABC-family, energy-dependent movement of polar lipids toward outer/luminal leaflet.
Scramblase	Bidirectional lipid scrambling; no ATP required. Plasma-membrane Ca ²⁺ -dependent scramblase can dissipate asymmetry.

Phosphatidylserine outside = signal. External PS marks apoptotic cells for phagocytosis and creates a procoagulant platelet surface.

Membrane transport + Na⁺/K⁺-ATPase

Transport types

Process	Protein?	Energy?	Direction / signature
Simple diffusion	No	No	Down gradient through bilayer; gases/lipophilic molecules cross most readily.
Facilitated diffusion	Yes	No	Down gradient; specific; protein-mediated; carrier transport can saturate and be competitively inhibited.
Primary active	Yes	Direct ATP	Can move solute against gradient; pumps.
Secondary active	Yes	Ion-gradient energy	Uses gradient made by primary transport to move another solute uphill.

Channel vs carrier

Channel: pore, very fast, down electrochemical gradient; may be voltage-, ligand-, or mechanically gated. CFTR is presented as an ATP-gated anion channel.

Carrier: binds solute + changes conformation; finite number -> saturation.

Secondary-active vocabulary

Symporter: transported particles move in the same direction.

Antiporter: transported particles move in opposite directions.

Na⁺-glucose cotransport: Na⁺ moves down its gradient while glucose is brought uphill.

Na⁺/K⁺-ATPase - memorize exactly

1 ATP -> 3 Na⁺ OUT + 2 K⁺ IN. The pump is electrogenic because more positive charge leaves than enters.

Gradient produced	Meaning
Outside -> inside Na ⁺ gradient	Na ⁺ strongly favors entry and can power secondary active transport.
Inside -> outside K ⁺ gradient	K ⁺ is concentrated intracellularly.
Inside-negative membrane potential	Electrical gradient contributes to cellular excitability.

What the gradients are used for

- **Secondary active transport** of nutrients/solutes.
- **Neuronal signaling** and depolarization.
- **Skeletal + cardiac muscle** function.

- **Enzyme regulation** through control of cytosolic Ca²⁺.
- **Na⁺/Ca²⁺ exchanger** helps keep intracellular Ca²⁺ low.
- **Hormone secretion**, including insulin release from beta cells.

Clinical anchors shown in the handout

Clue	Think
Improper lysosomal targeting / lysosomal enzyme defect	I-cell disease / Gaucher disease are examples named in the deck.
VLCFA/peroxisomal enzyme defect	X-linked adrenoleukodystrophy.
Peroxisomal protein-import defect	Zellweger syndrome.
Primary-cilium defect in renal tubular mechanosensing	Polycystic kidney disease.
Proteasome inhibition	Bortezomib as a cancer-therapy example.

If you see X -> think Y

- More prominent nucleoli -> **high ribosome biogenesis / protein synthesis.**
- Maternal inheritance + variable severity -> **mtDNA disease + heteroplasmy.**
- VLCFA problem -> **peroxisome.**
- Polyubiquitin tag -> **proteasome.**
- PS externalization -> **apoptotic clearance / procoagulant platelet surface.**
- Transport saturates -> **protein-mediated carrier process.**
- Direct ATP use -> **primary active transport**; gradient energy -> **secondary active transport.**

Last 30 seconds: nucleolus = ribosome factory; mitochondria = maternal mtDNA + heteroplasmy; lysosome = acid hydrolases; peroxisome = VLCFA + catalase; proteasome = ubiquitin; PS/PE/PI inside, PC/SM/glycolipids outside; 3 Na⁺ out / 2 K⁺ in / 1 ATP.

Built from the current Week 2 Cell Structure & Intracellular Organelles: An Overview handout and its official learning objectives.